Mass Gap for the O(N) Model at Large N via Brascamp-Lieb and Feynman-Kac

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Abstract

We outline a proof of the mass gap for the O(N) model in two dimensions at large N, with an exponentially small mass $m \sim \Lambda\, e^{-cN}$ arising from dimensional transmutation. The proof combines Brascamp-Lieb concentration of the σ-field, the Feynman-Kac representation for the φ-propagator, and a self-consistent gap equation that organizes all leading-order contributions. The $1/N$ corrections enter the *exponent* of the mass formula, making them multiplicative corrections to m^2 that are controlled for $N > N_0 \sim C\lambda v^2$.

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1 Setup

1.1 Model

O(N) LSM on $\Lambda \subset \mathbb{Z}^2$:

$$S[\varphi] = \tfrac{1}{2}\sum_i \langle \varphi^i, (-\Delta)\varphi^i\rangle + N\sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - v^2\big)^2, \quad v^2 = R^2/N,\ g^2 = N/R^2. \tag{1}$$

The LSM and NLSM share the same infrared physics; they differ at scale $M_\sigma \sim \sqrt{\lambda}\, v$ (the σ-field mass).

1.2 Hubbard-Stratonovich

With $\sigma(x) = |\varphi(x)|^2/N$, integrating out φ gives:

$$\langle \varphi^i(x)\varphi^i(0)\rangle = \mathbb{E}_\sigma\big[(-\Delta+\sigma)^{-1}(x,0)\big], \tag{2}$$

where $\mu_\sigma \propto e^{-N s_{\text{eff}}[\sigma]}$ with $s_{\text{eff}} = \frac{1}{2}\mathrm{Tr}\log(-\Delta+\sigma) + \sum_x \lambda(\sigma(x)-v^2)^2$.

2 Brascamp-Lieb concentration

2.1 Convexity at the bare minimum

The effective action s_{eff} has a minimum at $\sigma_0 = v^2$ (for λ above the convexity threshold). The Hessian at $\sigma_0 = v^2$ in Fourier is

$$H(k) = 2\lambda - \tfrac{1}{2}\Pi(k), \quad \Pi(k) = \int \frac{d^2p}{(2\pi)^2}\,\frac{1}{(p^2+v^2)((k-p)^2+v^2)}, \tag{3}$$

with $\Pi(0) = 1/(4\pi v^2)$. The convexity parameter is $\kappa = 2\lambda - \Pi(0)/2 > 0$ when $\lambda > 1/(16\pi v^2)$.

2.2 Cumulant bounds

Brascamp-Lieb for the log-concave measure $e^{-N s_{\text{eff}}}$ gives:

$$\mathrm{Cov}(\sigma(x),\sigma(y)) \le \tfrac{1}{N}\, H^{-1}_{xy}, \tag{4}$$

$$|\kappa_k(\sigma(x))| \le \frac{C^k\, k!}{(\kappa N)^{k-1}}, \quad k \ge 2. \tag{5}$$

The k-th cumulant bound follows from the sub-Gaussian concentration of log-concave measures (Borell's lemma).

3 Feynman-Kac and the gap equation

3.1 Path integral representation

From (**??**) and the Feynman-Kac formula:

$$\mathbb{E}_\sigma[G_\sigma(x,0)] = \int_0^\infty ds\, \mathbb{E}^W_{0\to x,s}\Big[\mathbb{E}_\sigma\big[e^{-\int_0^s \sigma(\omega(t))\,dt}\big]\Big]. \tag{6}$$

3.2 Cumulant expansion

Set $X = -\int_0^s \sigma(\omega(t))\,dt$. The σ-average is:

$$\mathbb{E}_\sigma[e^X] = \exp\Big(\sum_{k=1}^\infty \frac{\kappa_k(X)}{k!}\Big). \tag{7}$$

First cumulant: $\kappa_1(X) = -s\,\langle\sigma\rangle \approx -s\,v^2$.
Second cumulant:

$$\kappa_2(X) = \int_0^s\int_0^s \mathrm{Cov}\big(\sigma(\omega(t_1)),\sigma(\omega(t_2))\big)\,dt_1\,dt_2. \tag{8}$$

Using BL covariance and the Brownian bridge time integral:

$$\mathbb{E}^W[\kappa_2(X)] \approx \frac{s}{N}\int \frac{d^2k}{(2\pi)^2}\,\frac{H^{-1}(k)}{k^2/2} = \frac{2s}{N}\int \frac{d^2k}{(2\pi)^2}\,\frac{H^{-1}(k)}{k^2}. \tag{9}$$

The factor $1/k^2$ comes from $\mathbb{E}^W[|\int e^{ik\omega}\,dt|^2] \approx 4s/k^2$ (Brownian time integral in 2d).

3.3 The logarithmic divergence and dimensional transmutation

The integral in (**??**) diverges logarithmically in the IR:

$$\int \frac{d^2k}{(2\pi)^2}\,\frac{H^{-1}(k)}{k^2} \approx \frac{1}{2\pi\kappa}\int_{m_0}^\Lambda \frac{dk}{k} = \frac{1}{2\pi\kappa}\,\log\frac{\Lambda}{m_0}. \tag{10}$$

The IR cutoff m_0 is the physical mass (self-consistently determined). The effective mass from the first two cumulants:

$$m^2 = v^2 - \frac{1}{N\pi\kappa}\,\log\frac{\Lambda}{m_0}. \tag{11}$$

3.4 The self-consistent gap equation

Setting $m = m_0$ gives the **gap equation**:

$$\boxed{\log\frac{\Lambda^2}{m_0^2} = N\pi\kappa(v^2 - m_0^2) \approx N\pi\kappa v^2} \tag{12}$$

with solution

$$m_0^2 = \Lambda^2\, e^{-N\pi\kappa v^2} = \Lambda^2\, e^{-\pi\kappa R^2}. \tag{13}$$

This is positive for all parameter values — the mass gap exists. At fixed $g^2 = N/R^2$: $m_0^2 = \Lambda^2 e^{-\pi\kappa/g^2}$, independent of N (the 't Hooft scaling).

Remark 1. The gap equation (12) arises from the second cumulant alone. No multiscale analysis or cluster expansion is needed at leading order — just BL concentration + Feynman-Kac + self-consistency. The logarithmic IR divergence in 2d times the $1/N$ from BL produces an $O(1)$ self-energy, which is the mechanism of dimensional transmutation.

4 $1/N$ corrections

4.1 Corrections enter the exponent

The higher cumulants $\kappa_k(X)$, $k \geq 3$, contribute additional terms to the exponent. The mass formula becomes:

$$\log \frac{\Lambda^2}{m^2} = N\pi\kappa v^2 \Big(1 + \frac{c_1}{N} + \frac{c_2}{N^2} + \cdots\Big) \tag{14}$$

where $c_1, c_2, \ldots$ are computable from the higher cumulants and the σ-propagator. Crucially, the corrections are to the *exponent*, so

$$m^2 = m_0^2 \cdot e^{-\pi\kappa v^2 c_1 + O(1/N)} = m_0^2 \cdot (1 + O(\pi\kappa v^2/N)). \tag{15}$$

The corrections are **multiplicative** in m^2, with relative size $O(\kappa v^2/N)$. For $N \gg \kappa v^2 = \kappa R^2/N$, i.e., $N \gg \sqrt{\kappa R^2}$, the corrections are small and m^2 remains positive.

4.2 Bounding the k-th cumulant contribution

The k-th cumulant of $X = -\int \sigma(\omega)\, dt$ is bounded by:

$$|\kappa_k(X)| \leq s \cdot \frac{C^k\, k!}{(\kappa N)^{k-1}} \cdot \big(\log \tfrac{\Lambda}{m_0}\big)^{k-1} \tag{16}$$

using the BL cumulant bounds and the $k-1$ Brownian time integrations (each contributing a log-divergent factor). Since $\log(\Lambda/m_0) \approx N\pi\kappa v^2/2$:

$$\frac{|\kappa_k(X)|}{k!\, s} \leq \frac{1}{\kappa N}\Big(\frac{C\pi\kappa v^2}{2}\Big)^{k-1}. \tag{17}$$

The series $\sum_{k\geq 3} \kappa_k/(k!\, s)$ in the exponent converges when

$$\frac{C\pi\kappa v^2}{2} < 1, \qquad \text{i.e.,} \quad \kappa v^2 < \frac{2}{C\pi}. \tag{18}$$

This is a condition on the *coupling* (not on N), satisfied for moderate λ and v^2.

When this condition holds, the cumulant series converges and the correction to the exponent is:

$$\sum_{k=3}^{\infty} \frac{\kappa_k(X)}{k!\, s} = O\Big(\frac{(\kappa v^2)^2}{\kappa N}\Big) = O\Big(\frac{\kappa v^4}{N}\Big). \tag{19}$$

4.3 The mass gap theorem

Theorem 2 (Mass gap at large N)**.** *For the O(N) LSM on $\mathbb{Z}^2$ with $\kappa v^2 < 2/(C\pi)$ (a condition on the couplings), there exists $N_0 = O(\kappa v^4)$ such that for $N \geq N_0$:*

$$|\langle \varphi^i(x)\varphi^i(0)\rangle_c| \leq \bar{C}\, e^{-m|x|} \tag{20}$$

with $m > 0$ satisfying $m^2 = \Lambda^2 e^{-N\pi\kappa v^2(1+O(\kappa v^4/N))}$.

Proof outline.

1. **BL concentration:** The σ-measure $e^{-Ns_{\text{eff}}}$ is log-concave at $\sigma_0 = v^2$ with convexity κN. Cumulant bounds: $|\kappa_k| \leq C^k k!/(\kappa N)^{k-1}$.
2. **Feynman-Kac:** The φ-propagator is $\int_0^\infty ds\, \mathbb{E}^W[\mathbb{E}_\sigma[e^X]]$ with $X = -\int \sigma(\omega)\, dt$.
3. **Cumulant series:** $\mathbb{E}_\sigma[e^X] = e^{\sum \kappa_k/k!}$. The series in the exponent converges under the coupling condition.
4. **Gap equation:** The second cumulant gives the self-consistent equation (12) with solution $m_0^2 > 0$.
5. $1/N$ **control:** Higher cumulants correct the exponent by $O(\kappa v^4/N)$, a multiplicative correction to m^2. For $N > C\kappa v^4$: the correction is < 1 and $m^2 > 0$.
6. **Exponential decay:** The Feynman-Kac integral with $m^2 > 0$ gives $|G(x,0)| \leq Ce^{-m|x|}$ by standard estimates. □

5 Discussion

5.1 No cluster expansion or multiscale RG

At leading order, the gap equation emerges from a single self-consistency condition (the second Feynman-Kac cumulant), not from a multiscale decomposition. The BL cumulant bounds directly control the higher-order corrections. This is simpler than Kupiainen's cluster expansion [2], though both give the same result at large N.

5.2 The $N = 2$ obstruction

For $N = 2$, the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices in the angular field θ ($\pi_1(S^1) = \mathbb{Z}$). The σ-concentration cannot detect vortices. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no vortices, no BKT.

5.3 Comparison with Kupiainen

Both approaches use the Hubbard-Stratonovich transformation and $1/N$ as a small parameter. The differences:

	This work	Kupiainen [2]
σ-concentration	Brascamp-Lieb	Cluster expansion
φ-decay	Feynman-Kac	Random walk expansion
Gap equation	From FK cumulant	From saddle point
Model	LSM (finite λ)	NLSM ($\lambda = \infty$)

Since LSM and NLSM share the same IR physics, the results agree in the large-N limit.

References

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